KALINGA

Rationale Script for the Science Innovation Expo

KALINGA: A Low-Cost Baymax-Inspired Wearable Health Pendant with Vital-Sign Checks, Fall Alerts, Timed Pill Dispensing, and a Family Monitoring App

Length: about 3 minutes. **Speakers:** Researcher 1 (R1), Researcher 2 (R2), Researcher 3 (R3). **Props:** the KALINGA pendant on its cord, a phone with the KALINGA Monitor app open, placebo tablets in the carousel.

Time	Who	What to say	Show or do
0:00–0:15	R1	Good morning, honorable judges. We are [names] of [school], with our adviser, [adviser]. Our entry is KALINGA, a low-cost wearable health pendant that keeps watch over a loved one while the family is away.	Hold up the pendant.
0:15–0:45	R1	Every morning, many Filipino families leave a lolo or lola at home to go to work. In our community, [NEED: the family or barangay you observed and what happened]. Adults over 60 suffer the most fatal falls, and the longer they wait for help, the worse the harm (WHO, 2021). About half of patients on long-term medicines miss or mistime their doses (WHO, 2003).	Slide: problem in figures.
0:45–1:05	R2	Families try to cover this with a smartwatch, an alert pendant with a monthly fee, a pill dispenser, and a thermometer. Each does one job, and together they cost far more than most families can pay. No one device checks, reminds, and alerts the family at the same time.	Slide: Table 1, cost comparison.
1:05–1:35	R2	So we built KALINGA. Inspired by the soft healthcare companion of Big Hero 6, it is a 3D-printed pendant with an original two-dot face. It reads forehead temperature without touching, pulse rate and oxygen from a fingertip pad, senses falls and long stillness, has an SOS button, and turns a four-slot pill carousel to present each dose on time.	Point to each part on the pendant.
1:35–2:10	R3	Here is how it works. At eight o'clock the carousel opens the right slot, the pendant vibrates, and Lola tips out her tablet and presses OK. On my phone, the family sees “taken” at once. If she falls and does not cancel within ten seconds, a fall alert appears here, wherever the family is. The family can also tap Check now to ask for a new reading.	Live demo: dose cycle, then the app (demo mode: Simulate a fall).
2:10–2:35	R3	We are testing KALINGA against clinic instruments for accuracy, with simulated falls on mats, five days of doses, battery life, and usability with older adults and caregivers. [NEED: give the measured results once available.]	Slide: results tables.
2:35–2:50	R1	All the parts cost about PHP 2,150, with no monthly fee, and the shell can be printed and repaired locally. We are applying for an Industrial Design and a Utility Model with IPOPHL.	Slide: cost and IPOPHL.
2:50–3:00	R1	KALINGA means care. With it, no one has to be alone with a fall, a missed dose, or a fever. Thank you, and we welcome your questions.	All three face the judges.

Likely Questions and Short Answers

Is KALINGA a medical device? No. It screens, reminds, and alerts; it does not diagnose or treat. Readings are checked against clinic instruments, and abnormal results tell the family to call or see a health worker.

How accurate is it? We report the bias and 95% limits of agreement against reference instruments, and the SpO_2 Arms against the 4% ISO limit. [NEED: measured values.]

What if the older adult falls but is fine? The pendant beeps for 10 seconds; one press of OK cancels the alert.

What if there is no Wi-Fi? Records wait in the pendant and are sent when Wi-Fi returns; alerts need Wi-Fi. A 4G module is our next improvement.

How do you know the medicine was taken? The pendant records when the slot opened, whether it was tipped toward the palm, and when OK was pressed. It cannot prove swallowing, so we call it dose taken, late, or missed.

Why not just buy a smartwatch? A smartwatch does not time pills, many need a monthly fee for alerts, and it costs more. Table 1 compares the prices we canvassed.

Is the design copied from Big Hero 6? It is inspired by the idea of a soft, friendly healthcare companion. The shape and two-dot face are our own design, which we are filing as an Industrial Design.

How is the data protected? Only signed-in family accounts can read the data, under Firebase rules; ids are coded, following the Data Privacy Act of 2012.